

Stella Wan

Silicon Valley | Looking for SWE intern | Open to Relocation | 408-600-0591 | Stella.NEU.dev@gmail.com
[linkedin.com/in/mingzhuwan](https://www.linkedin.com/in/mingzhuwan) | github.com/Mingzhu666 | E-portfolio: mingzhu-journey.vercel.app

EDUCATION

Northeastern University: Master of Science in Computer Science | SV, CA Sep 2026 - Apr 2028
University of Tasmania: Master Of Information Technology And Systems | Australia Jan 2018 - Dec 2019
City of Hobart Student Ambassador, International Student Shine Awards, Vice-Chancellor's Leadership Award
Sanda University: Bachelor Of Information Science And Technology | China Sep 2008 - Dec 2012

TECHNICAL SKILLS

Languages & Frameworks: Java, Go, Python, C#, JavaScript, SQL, PyTorch, Spring, React.js, Next.js, Vue.js, Node.js, Angular
Technologies: AWS, GCP, Git, Jenkins, Docker, Kafka, Kubernetes, Zookeeper, ELK Stack, WebSocket, REST API, Swagger, RAG
Databases & Big Data: MySQL, PostgreSQL, Redis, MongoDB, Elasticsearch, Apache Spark, ETL, Redshift, Data warehouse

EXPERIENCE

Software Engineer | Australian Energy Market Operator | Sydney, AU Jun 2023 - May 2026

- Implemented backend services for utility account transfer registration workflows, supporting creation, retrieval, update, and status management of registration requests using **Spring Boot, JPA, and MySQL**.
- Adopted **Eureka, Spring Cloud Gateway** and **Ribbon** for service discovery/registration, traffic routing and load balancing, enabling **distributed deployment** to improve system reliability and scalability.
- Developed and deployed secure serverless user management services using **AWS Lambda**, integrating authentication and authorization mechanisms to ensure secure access to customer-facing APIs.
- Implemented a logging and observability platform with Log4js and the ELK stack (**Elasticsearch, Logstash, Kibana**), centralizing logs. Built **dashboards** for error counts, p95/p99 latency, request volume, service **metrics**, and configured **alarms**.
- Developed an enterprise web application using **React** and **TypeScript**, building reusable UI components, implementing client-side routing with React Router, and improving performance through **lazy loading**.
- Architected a robust testing framework using **Jest**, achieving 80% core component coverage. Worked with **DevOps** team to implement **Jenkins CI/CD** pipelines to automate test execution and reporting, enhancing quality assurance processes.

Software Engineer Intern | FNZ | Sydney, AU Mar 2022 - Feb 2023

- Improved investment strategy retrieval performance by implementing **Redis Cache-Aside** caching, achieving an 80% cache hit rate and significantly reducing repetitive **MySQL** queries for frequently accessed strategy data.
- Optimized **MySQL stored procedures** by rewriting inefficient **SQL**, eliminating joins, and tuning indexes, reducing query execution time and improving data processing efficiency.
- Deployed investment application with **Docker** image, utilizing **Kubernetes** to orchestrate containers and pods, deployed the service to multiple EC2 instances, and made the service open to the public with high availability.
- Executed **Test Driven Development** strategies to resolve critical UI rendering and core functionality bugs. These targeted fixes improved application stability, resulting in a significant reduction in user-reported issues.
- Introduced **GraphQL** to establish an efficient API layer for the financial investment interface, enhancing data retrieval efficiency and flexibility. Designed precise query structures and resolvers, enabling selective data fetching and reducing network payload.
- Refactored a legacy **C#** codebase by modularizing business logic, removing redundant code, and upgrading dependencies, improving maintainability and reducing technical debt.

PROJECTS

AI Knowledge Assistant on AWS | Python

- Built a serverless RAG-based knowledge assistant using **AWS Bedrock, Lambda, and Elasticsearch KNN** to support semantic retrieval over internal knowledge articles and natural-language Q&A. Designed backend APIs for ingestion, embedding, retrieval, and conversation workflows with metadata and chat context stored in **DynamoDB**, and integrated **CloudWatch logging and monitoring** for errors, latency, and API usage.

RedisLite | Go

- Built a **Redis-like in-memory key-value store in Go** with a custom RESP parser, Redis commands, and a goroutine-based TCP server for concurrent client connections. Implemented **AOF persistence, TTL expiration, and background cleanup workers** to keep maintenance work off the request path, improving responsiveness under load while balancing durability and memory usage.

CERTIFICATION

AWS Solution Architect, AWS Cloud Practitioner, Azure AI Fundamentals, Project Management Professional